

JUNIPERO VERBEKE

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Electrical Engineering M.S. candidate with a B.S. in Applied Physics, focused on **embedded systems and device physics**, with hands-on experience in **MATLAB/Simulink, Python, SPICE and TCAD** for research and production. Highly practical and passionate about semiconductor manufacturing: building and testing hardware for real environments in large-scale deployments. Experience in technical sales working with clients to implement low-voltage projects.

EDUCATION

M.S. Electrical Engineering — San Jose State University <i>GPA: 4.0</i>	Expected June 2027
B.S. Applied Physics — University of California, Santa Cruz <i>GPA: 3.5</i>	June 2025

PUBLICATIONS

A. Lu, **J. Verbeke**, P. Oldiges, R. Arghavani, H. Y. Wong, “Self-Heating and Radiation Hardness Studies of 3nm GAA-FET-Based SRAM with Different Substrate Isolation Techniques,” *arXiv:2607.05789*, 2026. [\[link\]](#)

EXPERIENCE

FinFET and Nanowire SRAM Radiation Hardness Studies TCAD	Oct. 2025 - Present
- Extensive usage of Synopsys Sentaurus TCAD Suite (SDevice, Structure Editor, SProcess, SVisual, Sentaurus Workbench, Silicon Workbench, Sentaurus Data Explorer). - Characterized and modeled FinFET and Gate-All-Around FET devices and SRAMs under radiation using TCAD simulation (device simulation, process simulation, layout)	
TSMC Washington Etch Equipment Engineer Intern	May 2026 – August 2026
- Programmed and maintained Applied Materials and Mattson etch tools. - Designed and fabricated custom troubleshooting devices and made modifications to tools using KiCAD, CAD tools and 3D printing. - Performed hardware troubleshooting and soldering on electronic, hydraulic, vacuum and cooling subsystems.	
Lawrence Livermore National Labs Research Collaboration	Nov. 2025 – March 2026
- Initiated a joint effort between LLNL & SJSU on response of a cable in a metal cavity under X-ray irradiation. - Implemented a QSPICE environment to analyze effects of radiation-induced electromagnetic fields.	
Veritas Managed Solutions, Inc. Applied Engineer	June 2021 – Present
- Spearheaded the formation of the engineering team, contributing to revenue growth from \$10M to \$28M through expansion into advanced low-voltage projects. - Engineered and estimated over 200 bids, with large-scale system deployments at Tesla Gigafactories NV & TX.	

RESEARCH PROJECTS

Experimental Verification of the Zeeman Effect Optics & Precision Alignment	Jan. 2025
- Designed an optical experiment to observe the splitting of mercury spectral lines in a magnetic field, utilizing a Fabry-Perot etalon, CCD camera, and polarization filters. - Performed precision optical alignment to project interference rings onto a sensor for data capture. - Analyzed spectral data to calculate the Bohr magneton.	
Compton Scattering & Electron Mass Determination Instrumentation	Data Analysis
Feb. 2025	
- Calibrated scintillation counters, photomultipliers, and multi-channel analyzers to detect gamma-ray scattering. - Validated the Klein-Nishina formula and Dirac theory by measuring scattered photon energy, performing Gaussian fitting and χ^2 error analysis on experimental data.	
Physics-Based Numerical Modeling of Dynamic Systems Python, MATLAB	Nov. 2023
- Developed numerical models for dynamic systems from Lagrangian mechanics to simulate physical behaviors. - Implemented RK4 and Euler algorithms in Python/NumPy to solve differential equations.	
Aerodynamic Analysis & Fluid System Validation Fluid Dynamics, XFLR5	Mar. 2025
Investigated fluid dynamic principles using Pitot/Venturi tubes and wind tunnels; acquired sensor data (pressure, velocity) to validate Bernoulli and Navier-Stokes concepts.	